

ASSIGNMENT 3 Report

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Part II– Concurrent Processes in Unix

Based on the execution of process 2b with the output file included in the submission, there was no deadlocks or livelocks experienced during any of the runs. This is because the semaphore and synchronization mechanisms ensured safe concurrent access to shared resources allowing multiple TAs to work concurrently while correctly processing all exams. The output showed that the TAs processed the exams in a non-deterministic order, meaning that the order in which exams were graded, and questions were completed varied from run to run.

Execution Order Observations

Concurrent activity:

Multiple TAs were observed working on the same exam at the same time. While one TA was checking or modifying the rubric for a particular question, another TA might simultaneously be working on a different question for the same exam. This parallelism allowed faster overall completion times but required careful coordination to avoid conflicts.

Rubric access and synchronization:

The program handled rubric modifications safely. When a TA modified a question's rubric, other TAs either checked the rubric or waited for the modification to complete before proceeding. This behavior ensured data integrity, preventing situations where two TAs might attempt to write conflicting changes simultaneously. This is an example of proper synchronization, likely implemented using locks, semaphores, or other concurrency control mechanisms.

Exam progression:

Each TA loaded exams sequentially from the available list. Once a TA completed all questions for a given student, it moved on to the next available exam. The TAs continued this process until all exams had been processed. This mechanism ensured that each exam was eventually marked without duplication or omission, while still allowing multiple TAs to work in parallel on different parts of the workload.

Completion and sentinel handling:

After finishing all real exams, each TA encountered the sentinel student ID 9999, which signaled that no further exams remained. Upon detecting the sentinel, each TA exited gracefully, ensuring that no TA remained active unnecessarily. The parent process monitored the TAs' completion and confirmed it with the message: *"All TAs finished. Parent exiting."* This final message indicates that all processes terminated successfully and that the concurrent marking system operated reliably throughout the execution.

Conclusion

The program demonstrates correct concurrent execution of multiple TAs marking exams. Randomized processing times and safe synchronization prevented deadlocks or livelocks, while all exams were correctly processed in a non-deterministic but orderly manner.